

Student Worksheet

Heating Processes & Ionising Radiation and Nuclear Reactions

NAME	CLASS	DATE

Heating Processes: claude.ai/code/artifact/d4b52c71-7b49-472e-b0bb-1252d2de17f9 Ionising Radiation & Nuclear Reactions: claude.ai/code/artifact/75c883c6-9b91-44ab-aea3-db4240254992

Part A — Heating Processes

A1. Heating Curve HEATING PROCESSES • MODULE 1

Set the substance to **Water**, mass **m = 0.50 kg**, heater power **P = 500 W**.

Q1. Read from the simulation: at what temperature does melting begin, and at what temperature does boiling begin?

Q2. Scrub to the moment melting ends. Record the elapsed time and energy supplied at that instant. Calculate ^[3] by hand the energy needed to melt 0.50 kg of ice using $Q = mL$ ($L_f = 334 \text{ kJ/kg}$ for water). Does your calculation agree with the simulation's reading?

Q3. Switch to Custom substance: melting point 200 °C, boiling point 300 °C, mass 1.00 kg, power 800 W. ^[3] Before pressing Play, predict which phase will take longest to heat through. Test your prediction, then explain using $Q = mc\Delta T$.

Q4. Using the particle view, explain why temperature does not rise during the melting or boiling plateaus ^[2] even though energy is still being supplied.

- Q5.** Scrub to partway through the melting plateau and record the temperature. Now decrease the heater power (e.g. from 500 W to 250 W) and record the temperature again. Using $Q = Pt$, explain why the temperature doesn't change even though less power is now being delivered — what quantity has the simulation kept fixed when you adjusted the power? [2]

A2. Thermal Equilibrium HEATING PROCESSES • MODULE 2

Body A: **Copper**, 0.20 kg, 95 °C. Body B: **Water**, 0.50 kg, 20 °C. ($c_{\text{copper}} = 385 \text{ J/kg}\cdot\text{K}$, $c_{\text{water}} = 4186 \text{ J/kg}\cdot\text{K}$)

- Q6.** Before clicking “Bring into contact”, calculate the predicted final temperature using $m_1c_1(T_f - T_1) = -m_2c_2(T_f - T_2)$. Show your working, then compare to the simulation's answer. [4]

- Q7.** Using the zeroth law of thermodynamics, explain why the two bodies stop exchanging net energy once they reach the same temperature. [2]

- Q8.** Without calculating, predict what happens to the final temperature if Body A's mass is made much larger than Body B's. Test this and describe what you see. [2]

A3. Black-Body Radiation HEATING PROCESSES • MODULE 3

Extension module — not part of this topic's official subject matter.

- Q9.** Set temperature to 310 K (body temperature), then 5778 K (the Sun). Record the peak wavelength for each and name the EM spectrum region each falls in. [2]

Extension — for students who finish early

- Q10.** Using Wien's Law, explain in one sentence why very hot objects glow blue-white while cooler ones glow red. [1]

Extension — for students who finish early

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- Q11.** Set surface area $A = 1 \text{ m}^2$ and record the "Radiated power" shown next to the spectrum. Increase A to 5 m^2 ($5\times$) and record it again — has it scaled by the same factor? Now check the peak wavelength at both areas. Using $P = \sigma AT^4$ and Wien's Law, explain why increasing surface area changes total power but not the object's colour or the shape of its spectrum. [3]

Extension — for students who finish early

A4. Heat transfer mechanisms NOT IN THE SIMULATION

- Q12.** The syllabus describes three mechanisms of heat transfer: conduction, convection and radiation. Give a one-sentence definition and one everyday example of each. Which one can transfer heat through the vacuum of space, and how do you know? [3]

Not shown in the simulation — use your own knowledge/research

A5. Mechanical work and efficiency NOT IN THE SIMULATION

- Q13.** Define efficiency, and give one everyday example where not all input energy becomes useful output — state where the "lost" energy goes. [2]

Not shown in the simulation — use your own knowledge/research

- Q14.** A heater supplies 800 J to a room, but only 650 J actually warms the air. Calculate the efficiency of this energy transfer. [2]

Not shown in the simulation — use your own knowledge/research

Part B — Ionising Radiation & Nuclear Reactions

B1. Radioactive Decay IONISING RADIATION • MODULE 1

Select **Iodine-131** ($T_{1/2} = 8.02$ days).

- Q15.** Using $N = N_0(1/2)^n$, calculate by hand what fraction remains after 3 whole half-lives. Scrub to 3.00 half-lives and check the "Fraction remaining" reading matches. [3]

- Q16.** Explain why the sample line (actual, random) doesn't exactly follow the smooth theoretical curve, and why the two would match more closely with a much larger sample. [2]

- Q17.** Estimate from the graph how many half-lives it takes for the sample to fall below 10% remaining. Convert this to real time for iodine-131. [2]

B2. Radiation & Matter IONISING RADIATION • MODULE 2

Use the reference table in the simulation to complete this table:

RADIATION	CHARGE	APPROX. MASS	IONISING POWER
Alpha (α)			
Beta (β)			
Gamma (γ)			

- Q18.** Switch to Penetration view. Test each shielding material (paper, aluminium, lead) for all three radiation types and record which are stopped and which pass through. [3]

- Q19.** Switch to Fields view, turn on the magnetic field. Using the charges above, explain why alpha and beta deflect in opposite directions, and why gamma travels straight. [3]

Q20. Which particle curves more sharply — alpha or beta? Using the charge and mass values above, explain why in terms of charge-to-mass ratio. [2]

B3. Nuclear Reactions IONISING RADIATION • MODULE 3

Select “Uranium-238 — alpha (α) decay”.

Q21. Write out the balanced nuclear equation shown. Add up mass numbers and atomic numbers on each side to confirm both are conserved. [3]

Q22. Switch to “Carbon-14 — beta-minus (β^-) decay”. Record Δm and E shown. Using the footer's conversions, show the working for $E = \Delta mc^2$ from Δm through to Joules and MeV. [4]

Q23. Switch to Fission & Fusion. Record the energy released for all six reactions shown (U-235 fission, Pu-239 fission, D-T fusion, He-3+He-3 fusion, p+ ^{11}B fusion, and the p-p chain). Which single reaction releases the most energy per event? Which releases the least? [3]

Q24. Two of the reactions above — He-3+He-3 and p+ ^{11}B — are described as “aneutronic”, meaning no free neutron is produced at any stage. Compare their equations to the D-T fusion equation and identify what's different. Suggest one practical reason a reactor designer might prefer an aneutronic reaction, even though it needs a much higher temperature to ignite. [2]

Binding energy per nucleon (not directly labelled in the simulation): divide E by the total nucleon count — 5 for D + T, 236 for U-235 + n.

- Q25.** Calculate the energy released per nucleon for D–T fusion and for U-235 fission. Which releases more per nucleon? This is why the syllabus describes fusion as releasing “more energy per nucleon” than fission. [4]

- Q26.** U-238 is radioactive; iron-56 is one of the most stable nuclides known. Using the strong nuclear force and the proton:neutron ratio, suggest one reason very large nuclei like U-238 tend to be unstable. [2]

Not shown in the simulation — use your own knowledge/research

- Q27.** Fission and fusion here must be deliberately induced, unlike the spontaneous decays in Module 1. [2]
Explain the difference between natural radioactive decay and artificial transmutation, using one example of each from the simulation.

Not shown in the simulation — use your own knowledge/research